# SAMYA SAMANTA

**SYNTHETIC ORGANIC CHEMIST | Ph.D. Candidate in Chemistry**

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## PROFESSIONAL SUMMARY

Ph.D. candidate in Chemistry with expertise in synthetic organic chemistry, small molecule synthesis, reaction optimization, and method development. Develops robust synthetic protocols by integrating mechanistic investigations with catalysis, electrochemistry, and photochemistry. Experienced in purification, analytical characterization, air-sensitive synthesis, technical documentation, and laboratory safety. Research includes first-author publications in JACS and Angewandte Chemie and mentoring six undergraduate and graduate researchers.

## TECHNICAL SKILLS

Synthetic chemistry: Organic synthesis; small molecule synthesis; retrosynthetic analysis; reaction optimization; method development; transition-metal catalysis; radical chemistry; electrochemistry; photochemistry; mechanistic investigations; air-sensitive synthesis.

Analytical characterization: Nuclear magnetic resonance (NMR); high-performance liquid chromatography (HPLC); gas chromatography-mass spectrometry (GC-MS); electron paramagnetic resonance (EPR); infrared spectroscopy (IR); UV-Vis spectroscopy; cyclic voltammetry; column chromatography.

Laboratory operations: Reaction setup; purification; compound characterization; experimental troubleshooting; protocol development; technical documentation; grant writing; laboratory safety; mentoring and training.

## RESEARCH EXPERIENCE

**Texas A&M University | Ph.D. Candidate, Chemistry | College Station, TX | 2021-2026**

- Developed synthetic methodologies for aziridine synthesis, beta-phenethylamine synthesis, and base-free C-N cross-electrophile coupling, resulting in first-author publications in JACS and Angewandte Chemie.
- Established a robust, scalable, and reproducible synthetic protocol for N-pyridinium aziridines, a bench-stable aziridine synthon, published in Organic Syntheses.
- Developed complementary aminyl radical generation platforms using redox-active N-pyridinium and N-phenothiazinium amine salts, expanding methods for N-centered radical generation.
- Integrated electrochemistry, photochemistry, transition-metal catalysis, and mechanistic studies to develop and optimize synthetic methodologies for C-N bond formation.
- Mentored six undergraduate and graduate researchers in synthetic organic chemistry, analytical characterization, air-sensitive techniques, and laboratory safety, enabling independent research.

## EDUCATION

Ph.D. in Chemistry, Texas A&M University | 2021-2026 (candidate) | GPA: 4.00/4.00
Thesis: Design and Application of Electrophilic Amine Salts for Modern Organic Synthesis
Advisor: David C. Powers

BS & MS in Chemistry, Indian Institute of Technology Kharagpur | 2015-2020 | GPA: 9.02/10.00

## ADDITIONAL RESEARCH EXPERIENCE

**Indian Institute of Technology Kharagpur | MS Researcher | India | 2019-2020**

- Performed extensive reaction optimization and substrate scope evaluation for organophotoredox-mediated amide and benzothiazole synthesis (Chemistry - A European Journal, 2020, 26, 3703-3708).

**Nanyang Technological University | Research Intern | Singapore | May-July 2019**

- Designed the synthesis and investigated the reactivity of osmium high-nuclearity carbonyl complexes for photoacoustic imaging (Journal of Cluster Science, 2021, 32, 929-935).

**University of Victoria | Research Intern | BC, Canada | May-July 2018**

- Developed synthetic routes to carbazole-bis(tetrazole) ligands and evaluated corresponding platinum and palladium complexes as sustainable oxidation catalysts (Journal of Coordination Chemistry, 2021, 74, 983-1008).

**Indian Institute of Technology Kanpur | Research Intern | India | May-July 2017**

- Synthesized a hexatopic triptycene hexa-acid linker and investigated metal-assisted self-assembly into honeycomb-structured metal-organic frameworks (MOFs).

## SELECTED PUBLICATIONS

Samanta, S.; Patel, L.; Manna, D.; Ojeda, M.; Powers, D. C. Base-Free C-N Cross-Electrophile Coupling. Journal of the American Chemical Society, 2026, accepted. Preprint: doi.org/10.26434/chemrxiv.15000151/v1.

Samanta, S.; Tan, H.; Chatterjee, S.; Powers, D. C. Synthesis of N-Pyridinium Aziridines. Organic Syntheses, 2026, 103, 78-100.

Samanta, S.; Ojeda, M.; Nguyen, P.; Powers, D. C. Reductive Activation of N-Phenothiazinium Amine Salts Affords N-Centered Radicals. Tetrahedron Letters, 2026, 183, 156204.

Samanta, S.; Biswas, P.; O'Bannon, B.; Powers, D. C. beta-Phenethylamine Synthesis: N-Pyridinium Aziridines as Latent Dual Electrophiles. Angewandte Chemie International Edition, 2024, 63, e202406335. Hot Paper; top viewed article in 2025.

## SCIENTIFIC COMMUNICATION AND LEADERSHIP

- Presented research at Pfizer Chemistry Connect (2024), the Southwest Regional Meeting of the American Chemical Society (2024), and the Gordon Research Conference on Organic Reactions and Processes (2025).
- Mentored prospective graduate students through the Texas A&M Graduate Student Association of Chemistry; served as a Phi Lambda Upsilon treasurer and judiciary board chair.
- Moderated an ACS Division of Chemical Health and Safety workshop (2023), introducing the RAMP framework through case studies and discussions of laboratory safety.

## SELECTED HONORS

Selected participant, 74th Lindau Nobel Laureate Meeting in Chemistry (2025); Derek and Christiane Barton Graduate Endowment (2025); Sharon Dabney Memorial Scholarship (2025); third place, Welch Conference poster presentation (2024); Grand Prize, TAMU Joint Research Symposium poster (2023).